

We want to prove that if

$$f_X(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

then $E(X) = \mu$

$X \sim N(\mu, \sigma^2)$ ✓

$Y = aX + b \longrightarrow X = (Y - b) / a$

To prove: $Y \sim N(a\mu + b, a^2\sigma^2)$

✓ $P(X \leq x) = P\left(\frac{Y-b}{a} \leq x\right)$

$= P(Y \leq ax + b)$ for $a > 0$

✓ $F_X(x) = F_Y(ax + b)$

✓ $f_X(x) = f_Y(ax + b) a$ by chain rule

✓ $f_Y(\underbrace{ax+b}_y) = \frac{1}{a} f_X(x) \quad \begin{matrix} y = ax + b \\ x = \frac{y-b}{a} \end{matrix}$

✓ $\frac{1}{a} f_X\left(\frac{y-b}{a}\right)$

$\frac{1}{a} \frac{e^{-\frac{(\frac{y-b}{a} - \mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$ transf. of RVs.

✓ $= \frac{1}{a\sigma\sqrt{2\pi}} e^{-\frac{(y - (b + a\mu))^2}{2\sigma^2 a^2}}$

$= N(a\mu + b, a^2\sigma^2)$

if $a < 0$

$$P(X \leq x) = P\left(\frac{Y-b}{a} \leq x\right)$$

$$= P(Y-b \geq ax)$$

$$= P(Y \geq b+ax)$$

$$F_X(x) = 1 - F_Y(b+ax)$$

$$f_X(x) = -f_Y(b+ax) \cdot a$$

$$= |a| f_Y(b+ax)$$

$$f_Y(b+ax) = \frac{1}{|a|} f_X(x)$$

CLT experiment

$$Y = \frac{X_1 + X_2 + \dots + X_n}{n}$$

where X_i = Coin toss with heads prob $\frac{1}{2}$
(independent)

$$E(Y) = \frac{1}{n} \times n E(X_1) = \frac{1}{n} \times n \times 0.5 = 0.5$$

$$\text{Var}(Y) = \frac{1}{n^2} \times n \text{Var}(X_1) = \frac{\text{Var}(X_1)}{n}$$